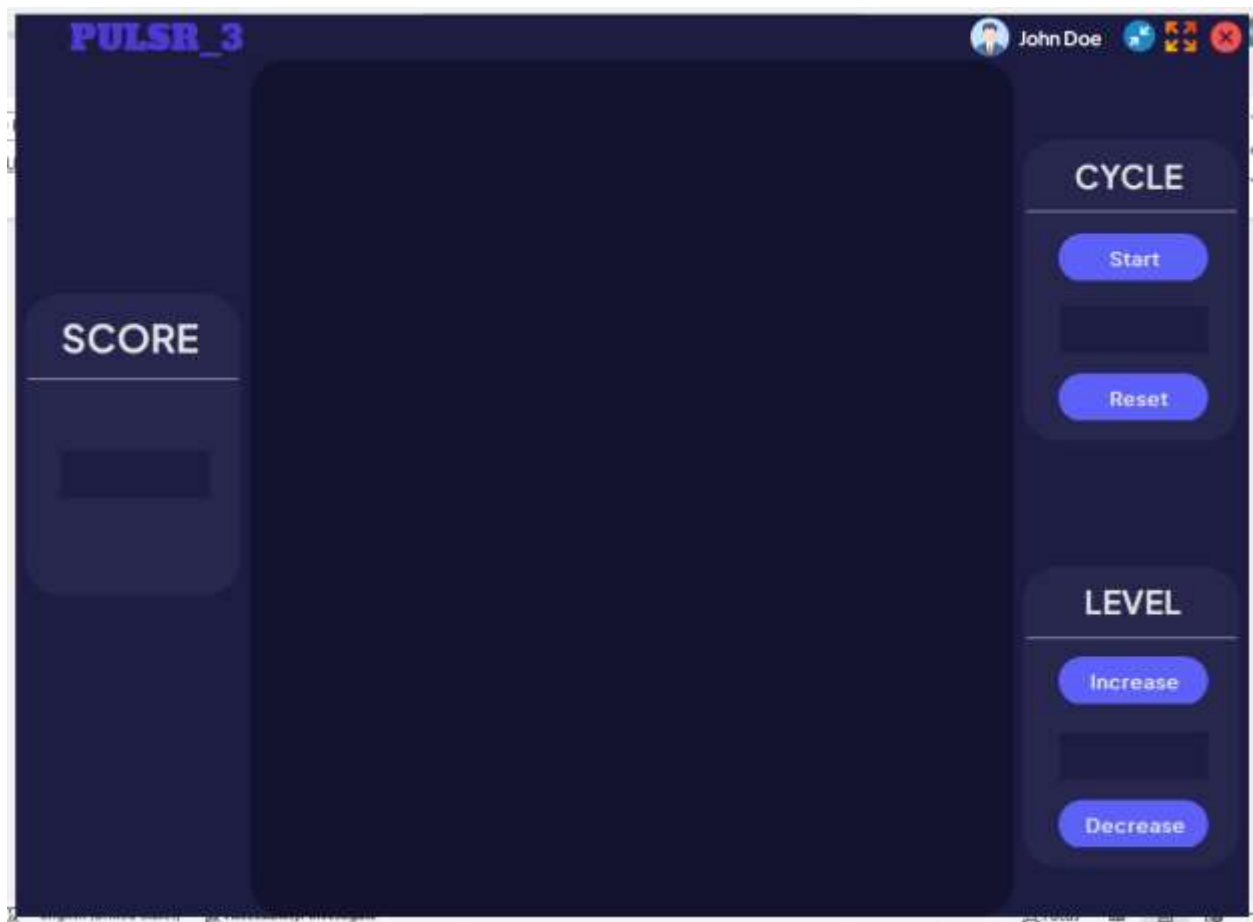


PULSR3 Therapy Robot Control Application Documentation.



M.O Opabode

Welcome to the PULSR3 Therapy Robot Control Application documentation! This guide provides comprehensive information on installing, configuring, and using the application.

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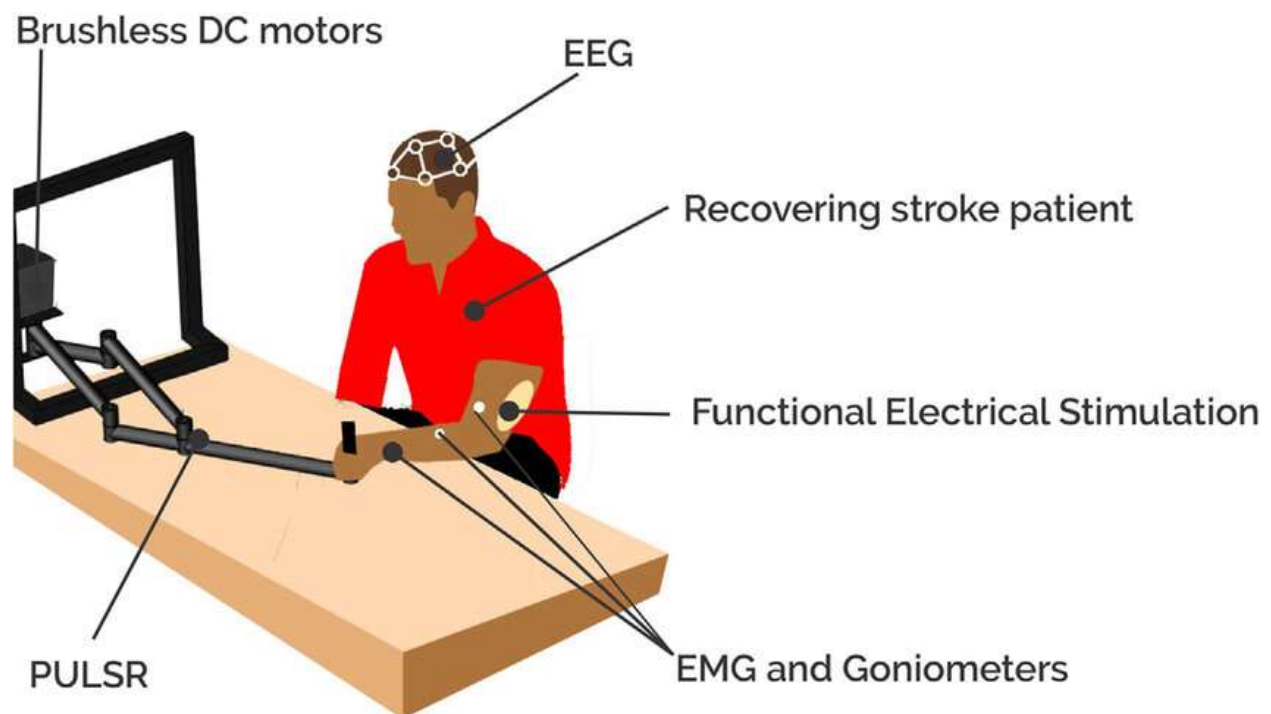
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OVERVIEW

Welcome to the Therapy Robot Control Application! This desktop application is designed to interface with the PULSR 2S/3 robot, allowing users to run therapy sessions and interact with the robot seamlessly. This is the first compiled application of the PULSR series robot.

Purpose:

The purpose of this application is to provide a user-friendly interface for controlling the PULSR 3 therapy robot. Users can initiate and manage therapy sessions, as well as interact with the robot to customize sessions based on individual needs.



The main components of the full PULSR system. PULSR, Platform for Upper Limb Stroke Rehabilitation.(Chapter 24 - Africa region: Nigeria)

INSTALLATION AND SETUP

Prerequisites:

Before running the application, ensure you have the following:

- PULSR (2s or 3) control box
- Batteries for the control box (24V Dc)

- USB cables
- Power supply for the control box

Installation Steps:

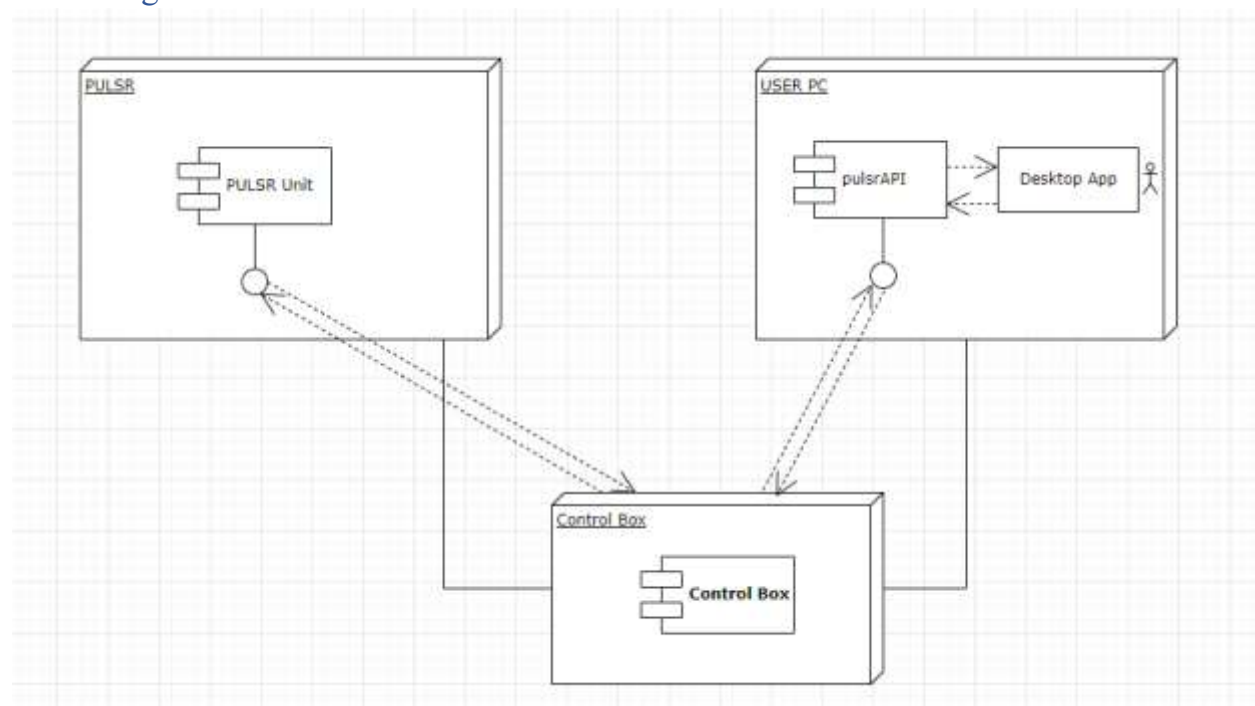
- Unzip the application package.
- Locate the executable (.exe) file.
- Locate the usb_ser(.txt) file.
- Locate the 'sessions_file' folder.
- Connect the control box to the PC using the USB cables and the control box to the robot.
- Connect the power supply (24 Dvc) to the control box.
- Ensure your workspace table is clean and ready for operation.

ARCHITECTURE

Components:

The main component of this application is the '**...API.cs**' file, which serves as the interface for interacting with the PULSR's robot via the control box. This file includes all the functions needed to communicate with the robot and serves as the API.

Flow Diagram:



Technologies Used:

- .NET framework
- C# programming language
- WinForms for the desktop user interface

CODE STRUCTURE

File Structure:

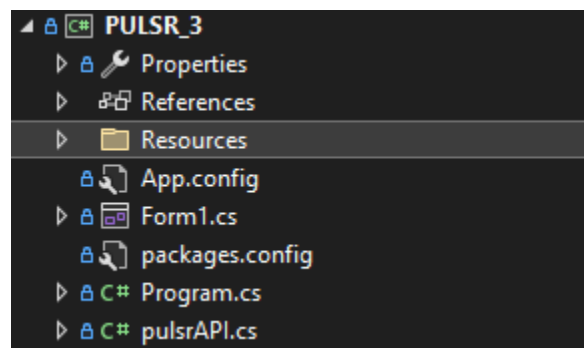
Sessions_files: Folder responsible for all session data.

usb_ser.txt: File responsible for COM port mapping.

Form1.cs: Main form for the user interface.

Program.cs: Entry point of the application.

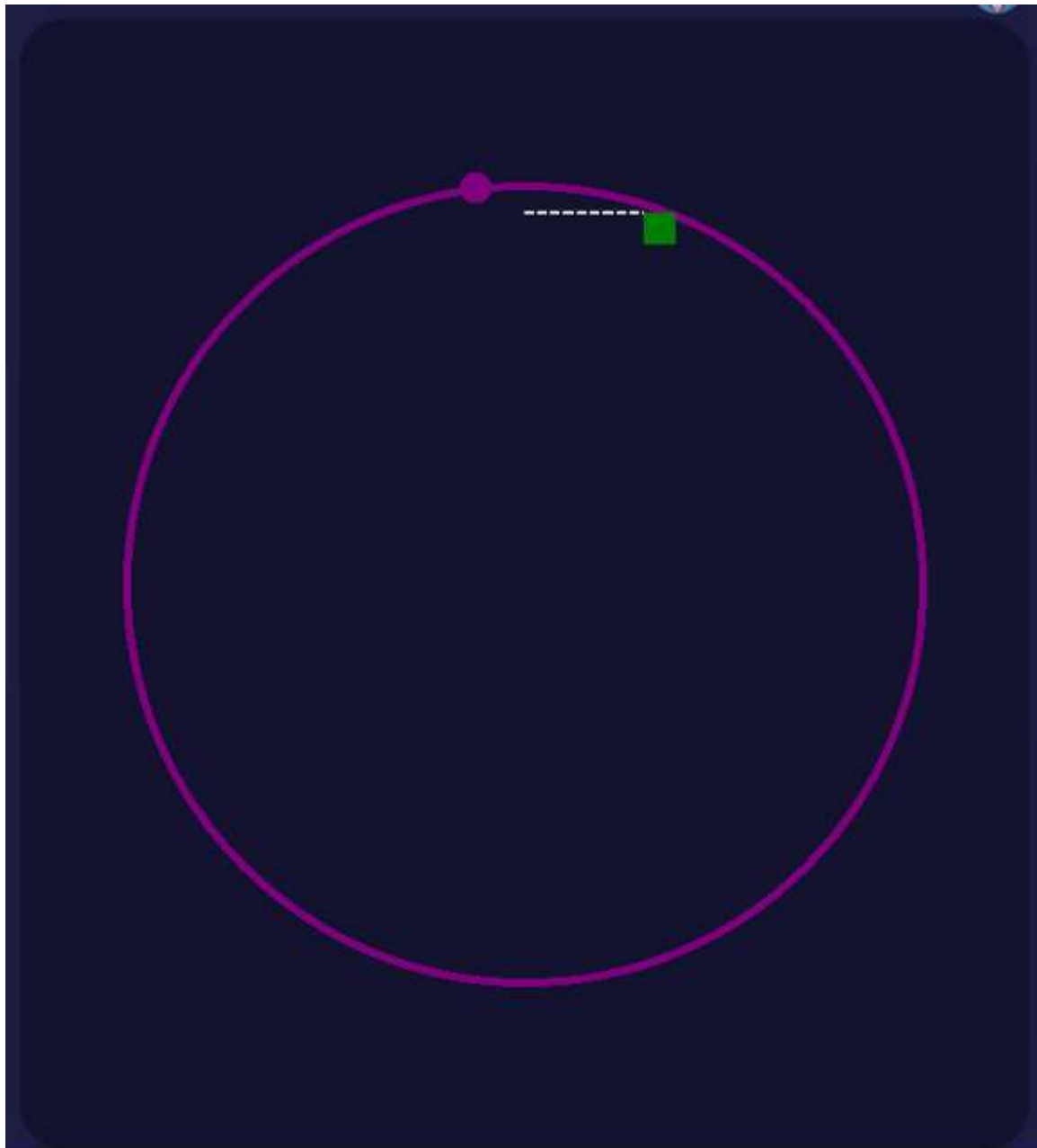
pulsrAPI.cs: File responsible for interfacing with the Pulsr3 robot.



USAGE

User Interface:

The main element of the user interface is the main screen where all the activities for learning the patients go on. Users interact by moving the end effector, which is shown on the main screen as a game to follow for motor function learning.



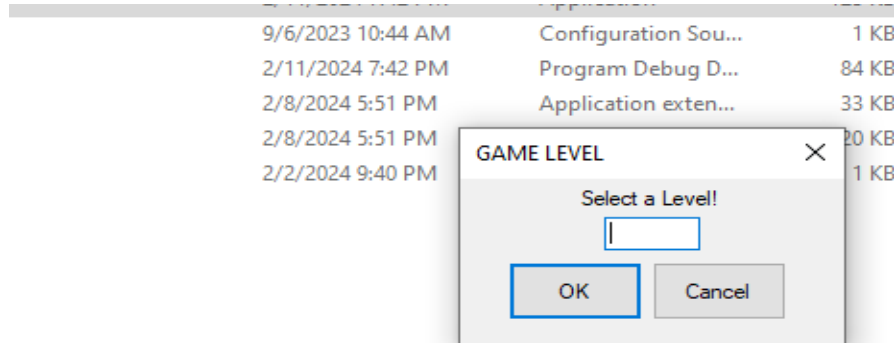
Running Therapy Sessions:

Technicians initiate therapy sessions for patients and guide them during the session. Data is collected at the end of the session.

CUSTOMIZATION

Game Modes:

At the start of the program, users are prompted to select a game mode: passive mode, assistive mode, or active mode, determined by the therapist.



Customizable Settings:

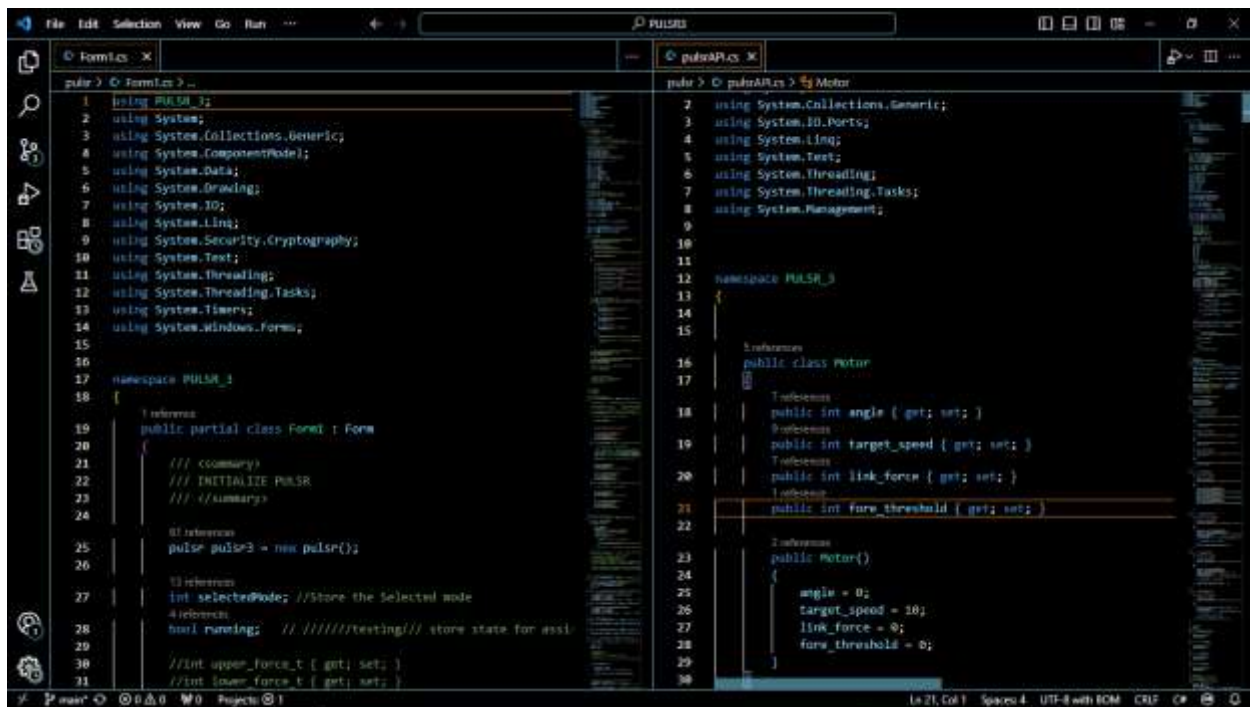
- Threshold (level).
- Speed.
- Sensitivity.
- Window scale.

These settings can all be customized to suit the user's needs.

INTEGRATION

Frontend and Backend:

The frontend of the application is the Form1.cs, where patients interact and receive feedback. All inputs and interactions are processed by the backend API, which communicates back and forth with the PULSR brain (control box).



ERROR HANDLING

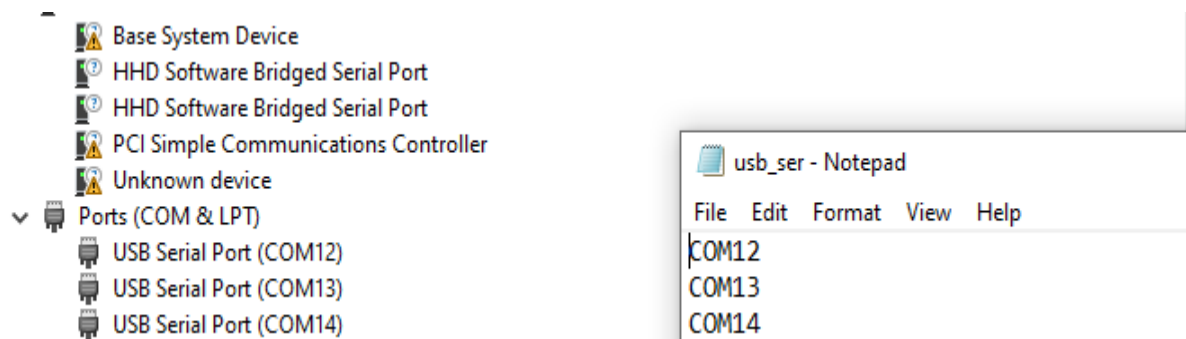
Common Errors:

- Application not opening
- Robot not starting
- Instruction and movement not in sync with the robot

Troubleshooting Tips:

Offer general troubleshooting tips for users to diagnose and resolve issues independently.

- Ensure **usb_ser.txt** contains the correct USB COM port name.



- Ensure it is tightly connected to the control box.
- Ensure the power supply is ON.

REFERENCES

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